

MedSegAgent: A Universal and Scalable Multi-Agent System for Instructive Medical Image Segmentation

Ziyan Huang, Haoyu Wang, Jin Ye, Yuanfeng Ji, Xiaowei Hu, *Member, IEEE*, Lihao Liu, Zhikai Yang, Wei Li, Ming Hu, Yanzhou Su, Tianbin Li, Yun Gu, *Member, IEEE*, Shaoting Zhang, *Senior Member, IEEE*, Yu Qiao, *Senior Member, IEEE*, Lixu Gu, *Senior Member, IEEE* and Junjun He

Abstract—Medical image segmentation is vital for clinical diagnosis and treatment; however, current solutions face three major limitations: (1) the lack of a universal framework capable of handling diverse modalities and anatomical targets, (2) the limited scalability to adapt to evolving clinical needs and new datasets, and (3) the lack of instructive interfaces that make models usable for non-expert users. To address these challenges, this paper presents MedSegAgent, a universal and scalable multi-agent system for instructive medical image segmentation. Specifically, MedSegAgent comprises five agents: one query parsing agent that processes natural language requests, three coarse-to-fine filtering agents (modality filtering, anatomical filtering, and label selection) for identifying relevant datasets and label values, and one execution agent responsible for model inference and result integration. Based on this framework, MedSegAgent utilizes 23 diverse datasets and pre-trained models to perform 343 types of segmentation across various modalities and anatomical targets. Experimental results demonstrate that MedSegAgent simplifies model selection while maintaining high performance, accurately identifying matching datasets and labels in 94.27% of queries and locating at least one suitable match in 99.03% of queries. MedSegAgent offers a universal and scalable solution for diverse medical image segmentation tasks, bridging the gap between user-friendly queries and the complexities of model selection and deployment. Our code is publicly available at <https://github.com/uni-medical/MedSegAgent>.

Index Terms—Universal Medical Image Segmentation, Multi-Agent System, Natural Language Instruction

Ziyan Huang, Haoyu Wang, Wei Li, Yun Gu, and Lixu Gu are with the Institute of Medical Robotics, Shanghai Jiao Tong University, Shanghai, China (e-mail: gulixu@sjtu.edu.cn).

Ziyan Huang, Jin Ye, Wei Li, Ming Hu, Yanzhou Su, Tianbin Li, Lihao Liu, Shaoting Zhang, Yu Qiao, and Junjun He are with the Shanghai Artificial Intelligence Laboratory, Shanghai, China (e-mail: hejunjun@pjlab.org.cn).

Jin Ye is with the Department of Data Science and AI, Faculty of IT, Monash University, Melbourne, Australia.

Xiaowei Hu is with the School of Future Technology, South China University of Technology, Guangzhou, China.

Yuanfeng Ji is with the Department of Radiation Oncology, Stanford University, Stanford, California, USA.

Zhikai Yang is with the Department of Biomedical Engineering and Health, KTH Royal Institute of Technology, Stockholm, Sweden.

Ming Hu is with the Department of Electrical and Computer Systems, Faculty of Engineering, Monash University, Australia.

Yanzhou Su is with the College of Computer and Data Science, Fuzhou University.

I. INTRODUCTION

MEDICAL image segmentation plays a crucial role in medical image analysis by enabling accurate delineation of anatomical structures and pathological regions, which is essential for diagnosis [1], treatment planning [2], and interventions [3]. In the past decade, the growth of specialized medical image datasets and advanced deep learning models [4], [5] have greatly improved segmentation accuracy, driving progress across a wide range of clinical applications.

However, these solutions often lack *generality* and focus on specific imaging modalities or anatomical targets. In practice, this limitation introduces several barriers to broader adoption. First, *model selection* is required for every application, requiring practitioners to determine which model—or combination of models—is most suitable for a specific clinical requirement. With an increasing number of specialized models, this decision-making process becomes more complex. Second, the *technical complexity* involved in deploying and maintaining multiple pipelines often requires specialized expertise, hindering seamless integration into routine clinical workflows. Third, these solutions operate in *isolation*, making it challenging to combine complementary models, even when such integration could enhance diagnostic insights. As a result, many advanced segmentation methods remain underutilized, limiting their potential to advance intelligent healthcare.

To address these challenges, researchers have pursued general-purpose solutions through two main strategies: the development of comprehensive datasets and the creation of general-purpose segmentation models. Efforts like TotalSegmentator [6], [7] have proven the feasibility of multi-structure annotation, offering broad coverage by annotating 104 whole-body structures in CT scans and later extending to 59 structures in MR imaging [7]. However, these datasets are constrained by limitations in sample size, data diversity, and annotation quality. On the other hand, general-purpose segmentation models like MedSAM utilize prompt-based interactive segmentation [8]–[10] to handle a variety of tasks. While these universal models are flexible, they demand significant computational resources and often exhibit lower accuracy and longer inference times compared to task-specific models with fewer parameters. Moreover, these approaches face a

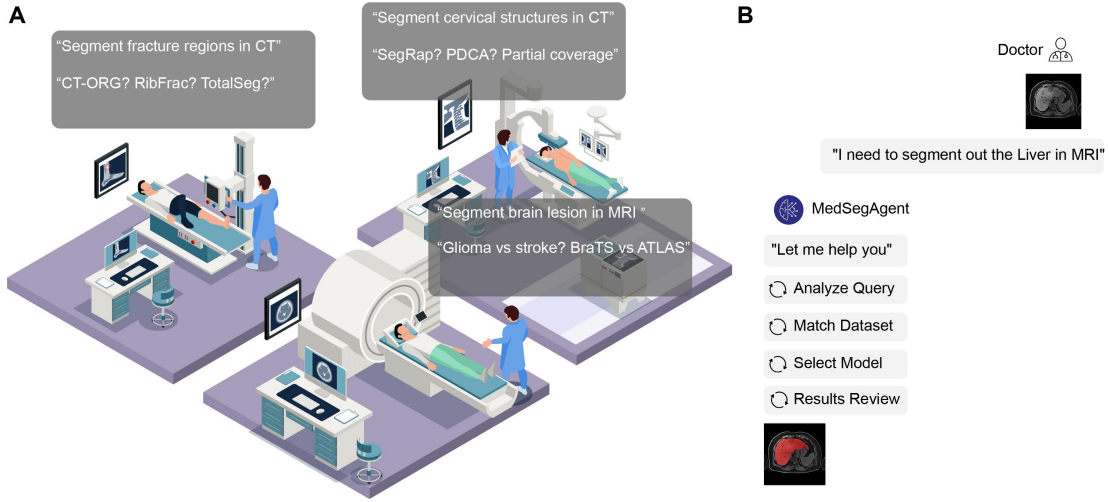


Fig. 1. Illustration of the clinical challenges and our proposed solution. A: Examples of clinicians struggling with model selection across different segmentation tasks. B: Our proposed MedSegAgent that bridges natural language requests and segmentation results.

scalability challenge as medical imaging evolves, with new protocols, disease targets, and clinical guidelines requiring extensive retraining or reannotation.

To fully leverage the strengths of each specialized segmentation model, we propose developing an intelligent management system capable of determining which models to use and how to apply them to diverse tasks. This system could potentially integrate Large Language Models (LLMs) or Multimodal Large Language Models (MLLMs) [11]–[13] as its core agents, enabling it to perform medical image segmentation based on textual instructions. To accomplish this, three key challenges must be addressed. First, translating clinical descriptions into a format that the system or model can understand and process is a significant hurdle. Additionally, the large number of medical image segmentation models and datasets, many of which overlap functionally, complicates the process of model selection. Finally, determining the optimal way to combine the results from multiple models is critical for achieving accurate and reliable segmentation outcomes.

In this paper, we develop MedSegAgent, a novel multi-agent system powered by LLMs/MLLMs, offering a universal framework for medical image segmentation based on user instructions. Specifically, MedSegAgent comprises five agents: a query parsing agent that converts natural language requests, three coarse-to-fine filtering agents (modality filtering, anatomical filtering, and label selection), and an execution agent responsible for model inference and result integration. This hierarchical architecture allows MedSegAgent to efficiently manage a vast repository of specialized datasets and their corresponding pre-trained models, enabling it to handle a broad range of segmentation tasks across various imaging modalities and anatomical regions.

We evaluated the MedSegAgent by integrating 23 diverse medical image segmentation datasets, each paired with pre-trained nnU-Net models. A benchmark of 343 queries was constructed to assess the system's ability to map natural language requests to relevant datasets and labels. Our results demonstrate that the proposed coarse-to-fine (C2F) strategy

outperforms other prompting methods. The system's scalability was verified through successful integration of five additional specialized datasets. Additionally, when compared with existing text-prompt-based general segmentation methods, our system not only demonstrates superior segmentation accuracy across multiple clinical scenarios but also maintains consistent performance across different textual descriptions of the same anatomical target, while other methods' performance can significantly degrade with such variations.

The main contributions of this work are as follows:

- We introduce **MedSegAgent**, a multi-agent system that achieves universal segmentation through orchestrating specialized models rather than training a single universal model.
- We leverage 23 diverse medical image segmentation datasets and their pre-trained models, enabling MedSegAgent to perform 343 types of segmentation across various modalities and anatomical regions.
- We establish a comprehensive benchmark of 343 natural language queries, demonstrating that MedSegAgent accurately identifies matching datasets and label values in **94.27%** of queries, and locates at least one suitable match in **99.03%** of queries.

II. RELATED WORK

A. Medical Image Segmentation

Medical image segmentation has advanced significantly through deep learning architectures [14]–[17] and large-scale datasets [2], [4], [6], [18]. The U-Net architecture [14] established the encoder-decoder paradigm with skip connections, while nnU-Net [15] demonstrates the power of automatic task-specific configuration, achieving strong performance across diverse benchmarks. Recent studies [19] further reveal that optimal architectures are highly task-specific, and appropriate model configuration can achieve the best trade-off between accuracy and efficiency for each task. However, such specialized models are typically designed for specific modalities or

anatomical targets, limiting their flexibility for broader clinical applications.

To address this limitation, researchers pursue two complementary strategies toward universal segmentation. The first builds comprehensive multi-category datasets: abdominal datasets [20]–[22] enable segmentation of over 10 organs in CT and MRI, SegRap2023 [2] provides annotations for 45 head-neck structures, and TotalSegmentator [6], [7] extends coverage to 104 CT and 59 MRI whole-body structures. The second strategy develops flexible model architectures. DoDNet [23] introduced dynamic segmentation heads with task embeddings, while Liu *et al.* [24] leveraged CLIP-driven text embeddings for semantic understanding. More recently, foundation models have emerged as a paradigm shift: MedSAM [8] demonstrates strong cross-modality generalization through large-scale training, SegVol [10] enables efficient 3D segmentation via zoom-out-zoom-in mechanism, and SAT [9] constructs an anatomical knowledge tree for text-prompted segmentation. Despite these advances, universal models still face challenges in matching specialized performance while managing computational demands and complex prompting protocols.

B. LLM-based Agents in Medicine

Large Language Models (LLMs) have demonstrated remarkable capabilities across medical applications, including clinical diagnosis [25], risk prediction [26], and report generation [27]. Beyond clinical settings, machine learning approaches have also shown value in broader healthcare decision-making, such as public health surveillance and predictive modeling [28]. To adapt LLMs for medical purposes, two main approaches exist: fine-tuning for task-specific optimization, which requires substantial computational resources and may reduce general capabilities, and inference-time enhancement strategies that preserve model strengths while enabling specialization.

Inference-time strategies have proven particularly effective. Retrieval-augmented generation (RAG) improves medical question answering and clinical guidance, with frameworks like Almanac [29] enhancing treatment recommendations and MIRAGE [30] establishing evaluation protocols. Multi-agent systems enable collaborative reasoning: MDAgents [31] facilitates adaptive collaboration for clinical diagnosis, while MedAgents [32] supports zero-shot medical reasoning through specialized agents. Tool integration extends LLM capabilities to practical applications, with AgentMD [26] automating risk prediction using clinical calculators, MMedAgent [33] orchestrating specialized tools for multi-modal diagnosis, and ChatCAD+ [34] combining CAD models with knowledge retrieval for medical image interpretation.

Unlike these approaches that augment LLMs with external medical tools, our work takes a complementary perspective: leveraging LLMs to intelligently orchestrate existing specialized segmentation models, thereby addressing the practical challenges of model selection and deployment in clinical workflows.

III. METHODOLOGY

A. Problem Definition

We consider a collection of medical image segmentation datasets, denoted as $\mathcal{D} = D_1, D_2, \dots, D_N$, where each dataset is defined as:

$$D_i = (\mathcal{M}_i, \mathcal{L}_i, f_i), \quad (1)$$

where

- $\mathcal{M}_i$: A pre-trained segmentation model associated with D_i ,
- $\mathcal{L}_i = \{l_1^i, l_2^i, \dots, l_{k_i}^i\}$: The set of anatomical or pathological labels, which may vary in terminology, granularity, and naming conventions,
- $f_i : \mathcal{L}_i \rightarrow \mathcal{P}(\mathbb{N})$: A mapping function that resolves synonyms and anatomical relationships to assign each label a distinct set of integer values used by the segmentation model.

Given a target medical image I and a free-form natural language query q , our objective is to identify all valid dataset-label pairs $\mathcal{C}$:

$$\mathcal{C} = \{(D, L) \mid D \in \mathcal{D}, L \subseteq \mathcal{L}_D, \text{satisfies}(I, q, D, L)\}, \quad (2)$$

where $\text{satisfies}(I, q, D, L)$ determines if D and L :

- 1) Match the imaging modality of I (e.g., CT, MRI, PET),
- 2) Fulfill the semantic requirements of q by aligning anatomical structures through terminological variations, laterality considerations, and hierarchical relationships using LLM-based reasoning.

Finally, the segmentation results $\mathcal{S}(I, q)$ are obtained by:

$$\mathcal{S}(I, q) = \{(D, L, \mathcal{M}_L) \mid (D, L) \in \mathcal{C}, \mathcal{M}_L = \mathcal{M}_D(I; f_D(L))\}, \quad (3)$$

where $\mathcal{M}_L$ is the segmentation mask from which we extract the regions corresponding to $f_D(L)$ to satisfy the query q .

B. System Overview

As shown in Fig. 2, MedSegAgent takes a natural language query and a medical image as input, and outputs segmentation masks for the requested anatomical structures. The system maintains a library of specialized segmentation models, each paired with dataset metadata (modality, anatomical labels). LLM-powered agents orchestrate the process: a *Query Parsing Agent* extracts the target modality and anatomy from the user's request; *Filtering Agents* progressively narrow down candidates through a coarse-to-fine (C2F) matching strategy; and an *Execution Agent* executes the selected models and produces final segmentation through MLLM-guided rank-aware ensemble.

C. Model Library and Scalability

MedSegAgent maintains a library of specialized segmentation models, each independently trained on a specific dataset using the nnU-Net [15] framework. Each model is paired with a standardized JSON metadata entry containing three fields:

- **Name**: A unique identifier for the dataset.
- **Modality**: The imaging modality or modalities (e.g., CT, MRI, PET).

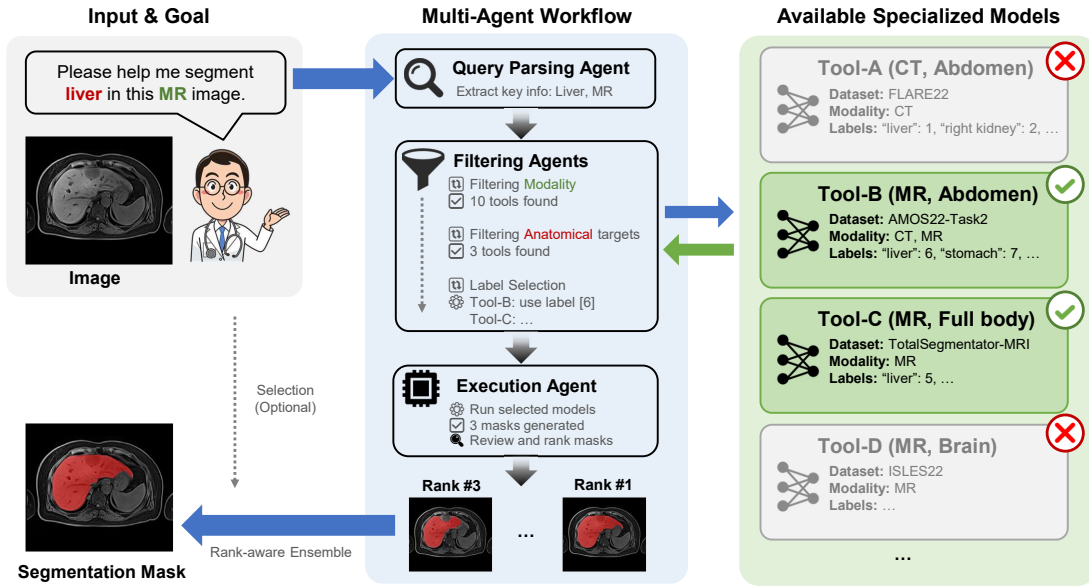


Fig. 2. Overview of the MedSegAgent framework. Given a natural language query and a medical image, the system identifies relevant models through coarse-to-fine matching and produces final segmentation via MLLM-guided rank-aware ensemble.

- **Labels:** A dictionary mapping anatomical targets to their corresponding integer label values.

To extend the system, one simply adds a new metadata entry and the corresponding pre-trained model.

D. Query Parsing and Coarse-to-Fine Matching

Algorithm 1 formalizes the coarse-to-fine matching procedure, which hierarchically narrows the candidate space (modality $\rightarrow$ anatomy $\rightarrow$ labels) to simplify LLM reasoning at each stage.

E. MLLM-guided Rank-Aware Ensemble

For each matched pair $(D, L) \in \mathcal{C}$, the Execution Agent runs the corresponding model to obtain segmentation outputs. When multiple models are matched, we employ a rank-aware integer voting ensemble. Let $\{P_1, \dots, P_K\}$ denote the K segmentation masks from matched models. To rank these outputs, we use a Multimodal LLM (MLLM) as follows. For each orthogonal view $v \in \{\text{axial}, \text{coronal}, \text{sagittal}\}$, we first compute the union mask and select the representative slice with the largest segmentation area:

$$z_v^* = \arg \max_z \left| \bigcup_{k=1}^K P_k^{(v,z)} \right|, \quad (4)$$

where $P_k^{(v,z)}$ denotes the 2D slice of mask P_k at position z along view v . For each z_v^* , the MLLM receives the original image slice and all model predictions, outputting a quality assessment a_v . The final ranking is obtained by aggregating the three view-specific assessments:

$$\text{rank}(P_1, \dots, P_K) = \text{MLLM}(a_{\text{axial}}, a_{\text{coronal}}, a_{\text{sagittal}}). \quad (5)$$

Given the ranked models, we assign deterministic integer weights based solely on rank: $w_r = K - r + 1$ for rank $r \in$

Algorithm 1 Query Parsing and Coarse-to-Fine Matching

- 1: **Input:** Natural language query q , dataset collection $\mathcal{D}$
- 2: **Output:** Matched dataset-label pairs $\mathcal{C}$
- 3: **Step 1: Query Parsing**
- 4: Extract modality m_q and anatomical target t_q from q using Query Parsing Agent.
- 5: **Step 2: Modality Filtering**

$$\mathcal{D}_{\text{mod}} = \{D \in \mathcal{D} \mid m_q \in \text{modality}(D)\}$$

- 6: **Step 3: Anatomical Filtering**
- 7: Use Anatomical Filtering Agent to select datasets containing the target anatomy:

$$\mathcal{D}_{\text{anat}} = \{D \in \mathcal{D}_{\text{mod}} \mid \exists l \in \mathcal{L}_D \text{ s.t. } l \text{ semantically matches } t_q\}$$

- 8: **Step 4: Label Selection**
- 9: Initialize $\mathcal{C} = \emptyset$.
- 10: **for** each $D \in \mathcal{D}_{\text{anat}}$ **do**
- 11: Use Label Selection Agent to determine $L \subseteq \mathcal{L}_D$ that semantically match t_q .
- 12: $\mathcal{C} \leftarrow \mathcal{C} \cup \{(D, L)\}$
- 13: **end for**
- 14: **return** $\mathcal{C}$

$\{1, \dots, K\}$. For each voxel x , we compute the weighted vote:

$$S(x) = \sum_{r=1}^K w_r \cdot b_r(x), \quad (6)$$

where $b_r(x) \in \{0, 1\}$ indicates whether the model at rank r predicts voxel x as foreground. The decision threshold is set to the weight of the second-ranked model: $\tau = w_2 = K - 1$. A voxel is accepted if and only if $S(x) > \tau$.

IV. EXPERIMENTS

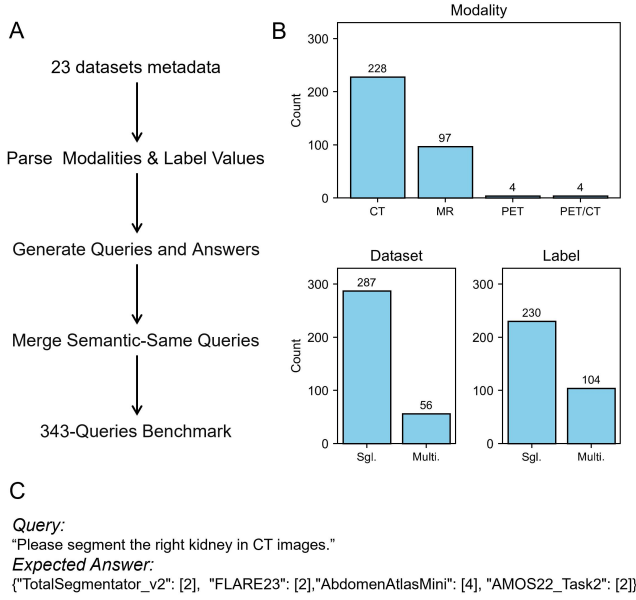


Fig. 3. Overview of the benchmark. A) Construction pipeline for generating the evaluation benchmark. B) Distribution of queries across modalities (top) and breakdown of single vs. multiple dataset/label queries (bottom). C) Example query-answer pair from the benchmark.

A. Datasets and Benchmark Construction

We integrated 23 diverse medical image segmentation datasets, each paired with a pre-trained nnU-Net model (Table I). To systematically evaluate our system’s ability to map natural language queries to the appropriate dataset-label pairs, we constructed a comprehensive benchmark through a structured three-step process (Fig. 3A):

Step 1: Metadata Collection. For each dataset, we extract supported modalities, anatomical label names, and their integer values. For example, FLARE22 supports CT with labels including “left kidney: 13” and “right kidney: 2”; AMOS22 supports both CT and MRI with “kidney_left: 3” and “kidney_right: 2”.

Step 2: Initial Query Generation. We iterate through each dataset, converting each (modality, label) pair into a query-answer pair. For example, FLARE22’s “left kidney” generates query “Please help me segment left kidney in CT” with answer {FLARE22: [13]}. For bilateral structures, we also generate composite queries: “Please help me segment kidney in CT” with answer {FLARE22: [2, 13]}.

Step 3: Semantic Merging. Queries with semantically equivalent targets are merged. Let $\text{lat}(x) \in \{L, R, B, \emptyset\}$ denote laterality (left, right, bilateral, unspecified), and $\text{children}(t)$ denote child structures of t . A label l from dataset D matches query target t_q if:

- *Semantic equivalence:* l refers to the same structure as t_q , satisfying $\text{lat}(t_q) = \emptyset \vee \text{lat}(l) \in \{\text{lat}(t_q), B\}$. For example, “Please help me segment left kidney in CT” merges FLARE22’s “left kidney: 13” and AMOS22’s

“kidney_left: 3” into {FLARE22: [13], AMOS22: [3]}; or

- *Hierarchical mapping:* $l \in \text{children}(t_q)$, requiring all children: $L_D^* = \text{children}(t_q) \cap \mathcal{L}_D$. For example, “Please help me segment kidney in CT” yields {FLARE22: [2, 13], AMOS22: [2, 3]}.

This process yielded 343 unique queries, each mapping to one or more dataset-label pairs (Fig. 3). The benchmark comprises 287 single-dataset and 56 multi-dataset queries, with 230 single-label and 104 multi-label queries (Fig. 3B). Error analysis on GPT-4o with standard prompting identified four failure modes with representative examples (Fig. 4): dataset missing (57.14%), dataset mismatch (15.58%), dataset incompleteness (9.09%), and label mismatch (18.18%).

B. Evaluation Metrics

1) Dataset and Label Selection Metrics: We define three complementary metrics: *Dataset Coverage (DC)* quantifies the fraction of datasets that are perfectly matched, providing an overall measure of accuracy. *Complete Coverage (CC)* verifies that no dataset is overlooked, thereby maximizing the utilization of available models for comprehensive analysis. *Minimal Coverage (MC)* ensures that at least one correct match is found, which can be sufficient in clinical workflows.

Formally, let

$$G = \{(d, L_d^G) \mid d \in D_G\}, \quad P = \{(d, L_d^P) \mid d \in D_P\}, \quad (7)$$

denote the ground-truth and predicted mappings for a given query, where d is a dataset, and L_d^G, L_d^P are sets of label values. Define $D_G = \text{keys}(G)$ and $D_P = \text{keys}(P)$.

Dataset Coverage (DC):

$$\text{DC}(G, P) = \frac{|\{d \in D_G \cup D_P \mid d \in D_P \text{ and } L_d^P = L_d^G\}|}{|D_G \cup D_P|}. \quad (8)$$

Complete Coverage (CC):

$$\text{CC}(G, P) = \begin{cases} 1 & \text{if } \text{DC}(G, P) = 1, \\ 0 & \text{otherwise.} \end{cases} \quad (9)$$

Minimal Coverage (MC):

$$\text{MC}(G, P) = \begin{cases} 1 & \text{if } \text{DC}(G, P) > 0, \\ 0 & \text{otherwise,} \end{cases} \quad (10)$$

2) Segmentation Performance Metrics: We employ two standard metrics to evaluate segmentation quality:

Dice Similarity Coefficient (DSC).

$$\text{DSC} = \frac{2|P \cap G|}{|P| + |G|}, \quad (11)$$

where P and G are the predicted and ground-truth segmentations, respectively.

Normalized Surface Dice (NSD).

$$\text{NSD}(G, P) = \frac{|B_{\partial G}(\tau) \cap \partial P| + |B_{\partial P}(\tau) \cap \partial G|}{|\partial G| + |\partial P|}, \quad (12)$$

where ∂G and ∂P denote the boundaries of the ground truth and the prediction, respectively. $B_{\partial G}(\tau)$ and $B_{\partial P}(\tau)$ represent border regions defined within a tolerance τ . In this work, we set the tolerance to 1 for all classes.

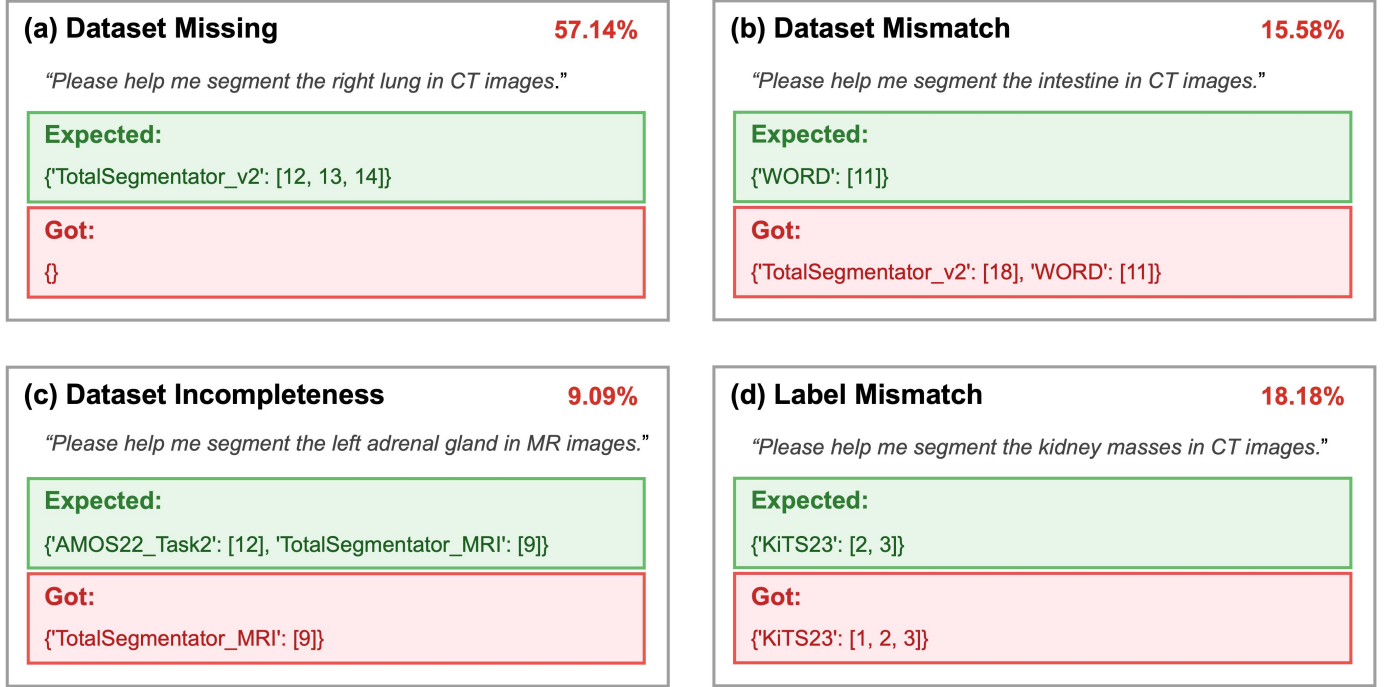


Fig. 4. Examples of common error types in dataset and label selection. Each panel shows a representative example comparing correct and incorrect responses. Green and pink backgrounds indicate correct and incorrect responses, respectively.

TABLE I
SUMMARY OF SELECTED MEDICAL IMAGE SEGMENTATION DATASETS. (NC: NON-CONTRAST, CE: CONTRAST-ENHANCED)

Dataset	Modalities	Body Region	Annotated Structures
TotalSegmentator v2 [6]	CT	Whole-body	117 structures including major organs (e.g., spleen, kidneys (L/R), liver), vessels (e.g., aorta, veins), bones, brain
TotalSegmentator MRI [7]	MRI	Whole-body	56 structures including major organs (e.g., spleen, kidneys (L/R), liver), major vessels (e.g., aorta, IVC), spine (vertebrae, intervertebral discs), muscle groups, brain
CT-ORG [35]	CT	Whole-body	liver, bladder, lungs, kidneys (L/R), bone, brain
AutoPET [36]	PET/CT	Whole-body	various tumor sites (e.g., melanoma, lymphoma, lung cancer)
SegRap2023 Task1 [2]	CT (NC, CE)	Head & Neck	45 structures including brain, brainstem, chiasm, temporal lobes (L/R), hippocampi (L/R), optic nerves (L/R), salivary glands (L/R), bony/cartilaginous landmarks
BraTS21 [37]	MRI (T1, T1Gd, T2, FLAIR)	Head & Neck	whole tumor, tumor core, enhancing tumor
ISLES22 [38]	MRI (ADC, DWI, FLAIR)	Head & Neck	stroke lesion
ISLES22 ATLAS [38]	MRI (T1)	Head & Neck	stroke lesion
Instance22 [39]	CT (NC)	Head & Neck	intracranial hemorrhage
HECKTOR2022 [40]	PET/CT	Head & Neck	GTVp, GTVnd
SegRap2023 Task2 [2]	CT (NC, CE)	Head & Neck	GTVp, GTVnd
MM-WHS [41]	MRI, CT	Heart	left/right atria, left/right ventricles, myocardium, ascending aorta, pulmonary artery
ACDC [42]	MRI	Heart	left ventricle, right ventricle, myocardium
ImageCAS [43]	CT	Heart	coronary artery
Parse22 [44]	CT (CE)	Thorax	pulmonary artery
ATM22 [18]	CT	Thorax	pulmonary airway
AbdomenAtlasMini [22]	CT	Abdomen	aorta, gallbladder, kidneys (L/R), liver, pancreas, postcava, spleen, stomach
AMOS22 Task2 [21]	MRI, CT	Abdomen	spleen, kidneys (L/R), gallbladder, esophagus, liver, stomach, aorta, IVC, pancreas, adrenal glands (L/R), duodenum, bladder, prostate or uterus
FLARE22 [20]	CT	Abdomen	liver, kidneys (L/R), spleen, pancreas, aorta, IVC, adrenal glands (L/R), gallbladder, esophagus, stomach, duodenum
WORD [45]	CT	Abdomen	liver, spleen, kidneys (L/R), stomach, gallbladder, esophagus, pancreas, duodenum, colon, intestine, adrenal, rectum, bladder, femurs (L/R)
KiTS23 [46]	CT	Abdomen	kidneys, renal tumors, renal cysts
LiTS [47]	CT	Abdomen	liver, liver tumor
Adrenal-ACC-Ki67-Seg [48]	CT (CE)	Abdomen	adrenocortical carcinoma

TABLE II
COMPARISON OF BASELINE METHODS ON A 343-QUERY BENCHMARK, USING GPT-4o-MINI AND GPT-4o BACKENDS.

Method	GPT-4o-mini [11]			GPT-4o [11]		
	DC (%)	CC (%)	MC (%)	DC (%)	CC (%)	MC (%)
Standard	11.50 $\pm$ 0.14	7.39 $\pm$ 0.17	15.35 $\pm$ 0.60	44.11 $\pm$ 4.19	39.46 $\pm$ 3.96	48.88 $\pm$ 4.27
CoT [49]	13.25 $\pm$ 0.46	9.23 $\pm$ 0.60	17.88 $\pm$ 0.61	72.37 $\pm$ 1.29	66.96 $\pm$ 0.94	77.74 $\pm$ 1.61
ReAct [50]	18.51 $\pm$ 1.74	13.02 $\pm$ 1.38	23.71 $\pm$ 1.60	76.86 $\pm$ 2.25	70.55 $\pm$ 2.39	82.22 $\pm$ 2.33
RAG [51]	22.85 $\pm$ 2.08	16.42 $\pm$ 2.26	28.86 $\pm$ 2.28	66.85 $\pm$ 1.74	59.77 $\pm$ 1.55	71.23 $\pm$ 1.44
Self Refine [52]	20.37 $\pm$ 1.33	14.67 $\pm$ 1.18	25.56 $\pm$ 0.94	75.04 $\pm$ 1.77	69.10 $\pm$ 1.16	80.17 $\pm$ 2.54
Multi Expert	55.96 $\pm$ 1.75	48.01 $\pm$ 2.15	64.43 $\pm$ 2.04	90.91 $\pm$ 0.18	85.42 $\pm$ 0.51	96.01 $\pm$ 0.84
C2F w/o Query Parsing	42.88 $\pm$ 0.83	37.90 $\pm$ 1.19	46.45 $\pm$ 0.96	91.97 $\pm$ 0.59	85.62 $\pm$ 0.27	96.60 $\pm$ 0.60
C2F w/o Modality Filter.	61.69 $\pm$ 0.33	54.81 $\pm$ 0.41	67.06 $\pm$ 0.24	93.22 $\pm$ 0.47	88.24 $\pm$ 0.36	97.08 $\pm$ 0.24
C2F w/o Anatomical Filter.	51.74 $\pm$ 1.77	45.68 $\pm$ 2.00	57.34 $\pm$ 1.17	91.01 $\pm$ 0.67	86.20 $\pm$ 0.84	95.34 $\pm$ 0.41
C2F (Ours)	67.04 $\pm$ 1.19	60.64 $\pm$ 1.33	71.91 $\pm$ 1.71	96.97 $\pm$ 0.12	94.27 $\pm$ 0.17	99.03 $\pm$ 0.17

TABLE III
COMPARISON OF ReAct VS. C2F ACROSS MULTIPLE LLM BACKENDS.

LLM Backend	ReAct [50]			C2F (Ours)		
	DC (%)	CC (%)	MC (%)	DC (%)	CC (%)	MC (%)
Claude-3.5-sonnet [12]	95.37 $\pm$ 0.49	91.84 $\pm$ 0.77	97.67 $\pm$ 0.00	96.60 $\pm$ 0.97	94.56 $\pm$ 1.18	98.06 $\pm$ 0.84
Claude-3.5-haiku [12]	76.95 $\pm$ 1.27	69.87 $\pm$ 1.22	83.87 $\pm$ 1.31	88.24 $\pm$ 1.98	82.51 $\pm$ 2.47	93.44 $\pm$ 2.27
Qwen2.5-72B [13]	66.45 $\pm$ 1.73	58.50 $\pm$ 2.35	74.05 $\pm$ 1.91	93.40 $\pm$ 1.30	88.53 $\pm$ 1.94	97.28 $\pm$ 1.38
Qwen2.5-32B [13]	70.83 $\pm$ 0.13	63.95 $\pm$ 0.44	77.06 $\pm$ 0.61	91.09 $\pm$ 0.35	85.71 $\pm$ 1.24	95.34 $\pm$ 0.41
Qwen2.5-14B [13]	67.20 $\pm$ 1.03	56.56 $\pm$ 0.83	77.56 $\pm$ 0.81	85.47 $\pm$ 0.59	79.35 $\pm$ 0.63	90.26 $\pm$ 0.41
Qwen2.5-7B [13]	27.76 $\pm$ 1.80	21.38 $\pm$ 1.18	35.76 $\pm$ 2.71	55.95 $\pm$ 5.14	47.23 $\pm$ 8.49	65.21 $\pm$ 2.19
Llama-3.1-70B [53]	64.46 $\pm$ 1.70	53.65 $\pm$ 1.27	75.22 $\pm$ 2.39	85.53 $\pm$ 4.36	78.91 $\pm$ 8.06	91.45 $\pm$ 1.18
Llama-3.1-8B [53]	21.43 $\pm$ 3.73	16.23 $\pm$ 2.87	30.13 $\pm$ 4.49	40.63 $\pm$ 1.58	33.92 $\pm$ 1.94	48.30 $\pm$ 1.10

C. Implementation Details

All experiments were conducted on a workstation equipped with an NVIDIA RTX 4090 GPU (24 GB VRAM), Intel Core i9-13900K processor, and 64 GB RAM. The multi-agent system was implemented using AutoGen 0.3 [54]. LLM inference was performed via official APIs.¹ Segmentation models were implemented using PyTorch 2.0 with the nnU-Net framework, following standard preprocessing pipelines (automatic resampling, intensity normalization, and cropping) without additional modifications. For LLM-based agents, we use temperature=0.7, top.p=1.0, and max.tokens=4096; all experiments were repeated 3 times independently with reported mean $\pm$ standard deviation to quantify inherent LLM variability. For local deployment experiments, we used 4-bit quantized Qwen2.5 models (7B and 14B) deployed via Ollama (v0.6.0). Inference latency was measured using the official FLARE22 [20] timing utility.

D. Comparison with Baseline Methods

To evaluate our coarse-to-fine (C2F) approach, we compare it against several prompting strategies using both GPT-4o and GPT-4o-mini as backends (Table II). We consider six baselines: **Standard** directly maps queries to dataset labels without explicit reasoning; **CoT** [49] employs chain-of-thought prompting; **ReAct** [50] alternates between reasoning and action phases; **RAG** [51] incorporates retrieval of dataset metadata before inference; **Self Refine** [52] adds a self-verification stage to refine predictions; **Multi Expert** is a

custom multi-agent system comprising specialized experts for different anatomical regions, with an assigner agent routing queries and a summary agent consolidating predictions.

Our C2F approach achieves the best performance on both backends, with 96.97% DC and 99.03% MC on GPT-4o. Compared to the strongest baseline (Multi Expert), C2F improves DC by over 6% on GPT-4o and 11% on GPT-4o-mini. Both methods reduce each agent’s scope, but C2F uses sequential filtering rather than parallel consultation. Notably, while RAG and Self Refine improve upon ReAct on GPT-4o-mini, they do not yield consistent gains on GPT-4o. Statistical significance was confirmed using two-sided Wilcoxon signed-rank tests on per-query accuracy across all 343 queries ($p < 0.001$ for all pairwise comparisons against baselines).

To further validate component contributions, we conducted ablation studies by removing individual agents (Table II, bottom rows). On GPT-4o-mini, removing Query Parsing causes the largest drop (−24.16% DC), while on GPT-4o, Anatomical Filtering is most critical (−5.96% DC). All ablations degrade MC, confirming that each agent contributes to the system’s overall effectiveness.

E. Generalizability to Other LLM Backends

We evaluate C2F against ReAct on open-source (Qwen2.5, Llama-3.1) and closed-source (Claude-3.5) models of various scales (Table III). C2F consistently outperforms ReAct across all backends, with larger gains on smaller models: Qwen2.5-72B improves by 26.95% DC, while Claude-3.5-sonnet gains only 1.23%. This suggests that C2F’s hierarchical decomposition particularly benefits models with limited capacity by

¹API versions used: gpt-4o-2024-08-06, gpt-4o-mini-2024-07-18, claude-3.5-sonnet-20241022, claude-3.5-haiku-20241022.

TABLE IV
PERFORMANCE COMPARISON USING DIFFERENT QUERY VARIANTS.

Query Variants	DC (%)	CC (%)	MC (%)
Standard	96.97 $\pm$ 0.12	94.27 $\pm$ 0.17	99.03 $\pm$ 0.17
Synonyms	91.02 $\pm$ 1.69	87.17 $\pm$ 2.27	94.46 $\pm$ 1.45
Δ vs. Standard	$\downarrow$ 5.95	$\downarrow$ 7.10	$\downarrow$ 4.57
Example failure: “hystera” cannot be mapped to “uterus”			
Abbreviations	92.35 $\pm$ 0.81	88.34 $\pm$ 1.26	95.24 $\pm$ 0.60
Δ vs. Standard	$\downarrow$ 4.62	$\downarrow$ 5.93	$\downarrow$ 3.79
Example failure: “OD” (oculus dexter) cannot be mapped to “right eye”			

breaking complex matching into simpler sub-tasks. Notably, smaller models with C2F can surpass larger models with ReAct: Qwen2.5-14B + C2F (85.47%) outperforms Qwen2.5-72B + ReAct (66.45%), demonstrating that C2F effectively compensates for model capacity limitations.

Among open-source alternatives, Qwen2.5-32B with C2F achieves 91.09% DC, approaching GPT-4o performance (96.97%) while remaining locally deployable, offering a practical option for privacy-sensitive clinical environments.

F. Robustness to Query Variations

To evaluate robustness to different query formulations, we manually constructed two additional test sets by rephrasing all 343 queries using uncommon synonyms (e.g., “renal” for “kidney”) and medical abbreviations (e.g., “IVC” for “inferior vena cava”). As shown in Table IV, MedSegAgent maintains strong performance across query variants. Synonyms and abbreviations cause modest DC drops of 5.95% and 4.62%, respectively, while MC remains above 94%. Failures occur mainly with very rare terms (e.g., “hystera” for uterus) or ambiguous abbreviations (e.g., “OD” for right eye). These results confirm that the LLM’s semantic understanding effectively handles common query variations in clinical practice.

TABLE V
PERFORMANCE BEFORE AND AFTER SYSTEM EXTENSION.

Query Set	DC (%)	CC (%)	MC (%)
Original (343)	96.97 $\pm$ 0.12	94.27 $\pm$ 0.17	99.03 $\pm$ 0.17
After Ext. (343)	96.87 $\pm$ 0.13	94.17 $\pm$ 0.29	98.93 $\pm$ 0.17
New (10)	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00	100.00 $\pm$ 0.00

G. Scalability Evaluation

To evaluate scalability, we integrated five additional datasets covering new modalities and targets: LNQ2023 [55], FUMPE [56], TDSC-ABUS2023, SPPIN [57], and Cross-MoDA2023 [58]. As shown in Table V, performance on the original 343 queries remains stable after extension (DC: 96.97% $\rightarrow$ 96.87%), while new queries targeting the added datasets achieve 100% accuracy. This confirms that MedSegAgent can incorporate new models without degrading existing capabilities.

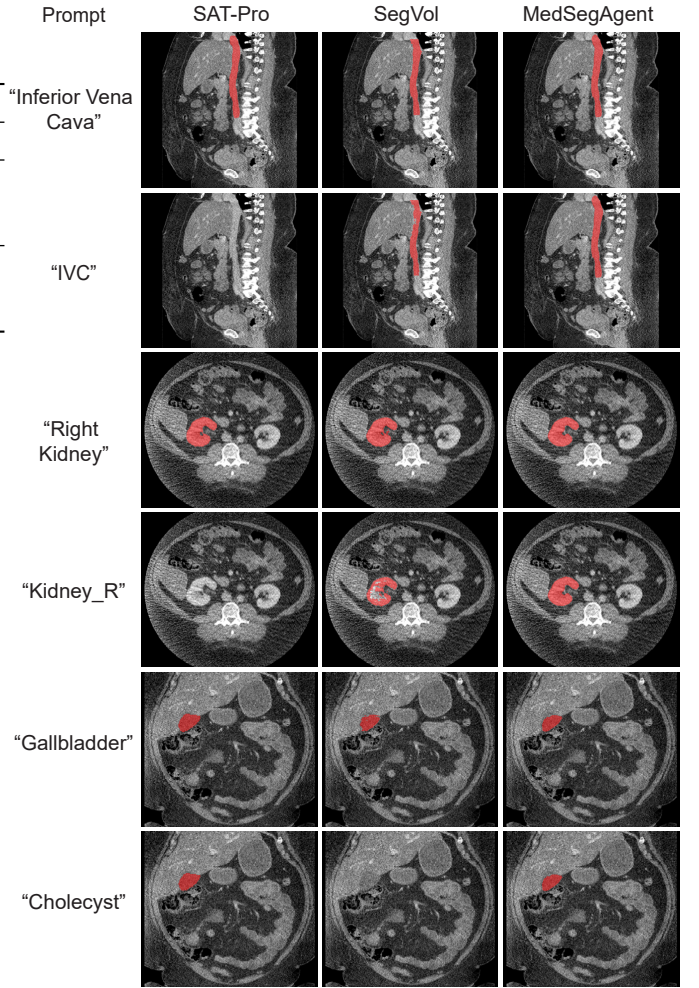


Fig. 5. Visual comparison of segmentation results using different text prompts across SAT-Pro, SegVol, and MedSegAgent (ours).

H. Comparison with General-Purpose Medical Image Segmentation Methods

We compare MedSegAgent with state-of-the-art general-purpose segmentation models: TotalSegmentator [6], MedSAM2 [59], SAM-Med3D [60], SegVol [10], and SAT-Pro [9]. To ensure fairness, we use a single pre-trained nnU-Net per dataset in MedSegAgent, without ensembling. Experiments were conducted on FLARE22 [20] (abdominal CT, 13 organs, 50:50 train-test split), ACDC [42] (cardiac MRI, 100:50 patient split), and AutoPET [36] (PET/CT tumor, 398:103 split).

As shown in Table VI, MedSegAgent achieves the highest DSC across all three datasets: 86.91% on FLARE22, 91.53% on ACDC, and 76.97% on AutoPET. All pairwise differences between MedSegAgent and baseline methods are statistically significant ($p < 0.001$, two-sided Wilcoxon signed-rank test). TotalSegmentator performs competitively on FLARE22 (85.01%) but is limited to CT. SegVol, primarily trained on CT, fails on MRI-based ACDC (0.94% DSC). SAT-Pro shows broader adaptability but still underperforms MedSegAgent. Figure 5 further demonstrates that MedSegAgent maintains consistent results across different text prompts, while SAT-Pro and SegVol exhibit significant variability.

TABLE VI
PERFORMANCE COMPARISON OF GENERAL-PURPOSE MEDICAL IMAGE SEGMENTATION METHODS ACROSS DIFFERENT DATASETS.

Method	Prompt	FLARE22 [20]		ACDC [42]		AutoPET [36]	
		DSC (%)	NSD (%)	DSC (%)	NSD (%)	DSC (%)	NSD (%)
TotalSegmentator [6]	-	85.01 $\pm$ 16.78	75.81 $\pm$ 19.08	-	-	-	-
MedSAM2 [59]	BBox	56.51 $\pm$ 38.44	52.07 $\pm$ 34.80	32.77 $\pm$ 30.28	34.53 $\pm$ 16.22	9.93 $\pm$ 16.22	4.10 $\pm$ 6.71
SAM-Med3D [60]	Point	50.84 $\pm$ 30.29	30.95 $\pm$ 20.92	44.42 $\pm$ 26.20	32.56 $\pm$ 11.55	17.12 $\pm$ 24.86	9.84 $\pm$ 14.64
SegVol [10]	Text	81.16 $\pm$ 14.57	69.81 $\pm$ 16.92	0.94 $\pm$ 2.88	0.61 $\pm$ 1.71	0.20 $\pm$ 1.02	0.21 $\pm$ 0.91
SegVol [10]	Text + BBox	84.44 $\pm$ 8.92	71.70 $\pm$ 14.12	0.41 $\pm$ 2.96	0.33 $\pm$ 2.03	17.56 $\pm$ 22.75	8.02 $\pm$ 10.04
SAT-Pro [9]	Text	81.39 $\pm$ 14.73	69.65 $\pm$ 17.05	87.36 $\pm$ 6.29	72.49 $\pm$ 8.01	71.30 $\pm$ 20.78	50.79 $\pm$ 17.47
MedSegAgent (Ours)	Text	86.91 $\pm$ 15.76	81.89 $\pm$ 17.84	91.53 $\pm$ 4.54	82.00 $\pm$ 7.05	76.97 $\pm$ 15.14	58.85 $\pm$ 17.25

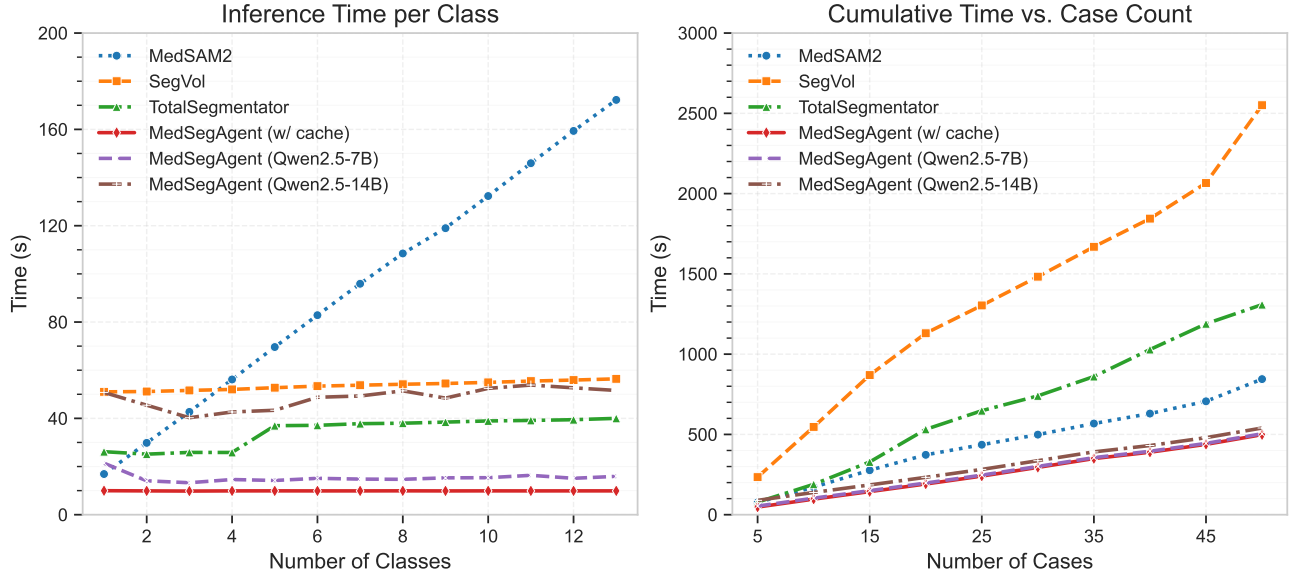


Fig. 6. Inference latency on FLARE22. **Left:** time to segment 1–13 organs in a single volume. **Right:** cumulative time to segment one organ over 5–50 volumes. MedSegAgent is shown in three configurations: cached mode, Qwen2.5-7B, and Qwen2.5-14B. Lower is faster.

I. Inference Efficiency

Inference latency was measured on the FLARE22 official validation set, timing each case end-to-end. SAT-Pro was excluded because it cannot run on a single RTX 4090. We compared MedSegAgent against SegVol, MedSAM2, and TotalSegmentator. We evaluated MedSegAgent under two scenarios: (1) *warm start* (cached mode), where repeat prompts reuse the previously selected model with no selection overhead; and (2) *cold start* (uncached mode), where model selection is performed via quantized Qwen2.5 models, adding approximately 5s (7B) or 40s (14B) of initial latency. We assessed performance in two typical scenarios: (1) segmenting 1–13 organs in a single volume and (2) segmenting a fixed organ across 5–50 volumes.

Figure 6 summarizes the results. In the multi-class test on a single volume, SegVol and MedSAM2, which require a separate prompt per organ, show near-linear latency growth. TotalSegmentator exhibits discrete runtime jumps when loading extra expert models. By contrast, the FLARE22 champion model—measured for inference time only—maintains a nearly constant per-case segmentation time below 12 s. MedSegAgent’s uncached quantized Qwen2.5 variant adds about 40 s overhead with the 14B model and about 5 s with the 7B model.

In the single-class test across multiple volumes, only an initial model-selection step is required; subsequent inferences reuse the selected model at constant per-case latency.

J. MLLM-guided Rank-Aware Ensemble Evaluation

To evaluate the rank-aware ensemble strategy, we selected ten anatomical structures from BTCV [61] (following the organ selection in [62]) that are present in all four datasets: FLARE22 [20], AMOS22 [21], TotalSegmentator [6], and WORD [45]. Dataset-specific nnU-Net models were trained and tested on 30 BTCV cases (no overlap with training), comparing oracle bounds (Best/Worst Selection), Random Selection, Majority Voting Ensemble, and our MLLM-guided methods (top-model Selection and rank-aware Ensemble) using GPT-4o [11].

As shown in Table VII, individual models achieved average DSC ranging from 80.10% (WORD) to 84.05% (AMOS22), with complementary organ-specific strengths: TotalSegmentator excelled in liver (96.13%), while FLARE22 performed best on adrenal glands (75.20%/74.39%). The oracle bounds (Best: 86.86%, Worst: 77.17%) demonstrate substantial room for intelligent selection. Figure 7 illustrates the case-by-case variability across results integration strategies. Among automatic

TABLE VII
SEGMENTATION PERFORMANCE ON BTCV DATASET ACROSS DIFFERENT MODELS FOR DICE SIMILARITY COEFFICIENT (DSC).

Model	Spleen	Kidney (R)	Kidney (L)	Gallbladder	Liver	Stomach	Pancreas	Adrenal (R)	Adrenal (L)	Esophagus	Average
FLARE22 [20]	90.50 ± 15.42	86.93 ± 15.91	86.96 ± 19.06	74.13 ± 29.60	94.59 ± 7.15	88.66 ± 11.69	81.96 ± 6.40	75.20 ± 4.62	74.39 ± 16.56	79.88 ± 17.19	83.32 ± 8.64
AMOS22 [21]	93.36 ± 13.30	90.61 ± 15.71	89.44 ± 17.66	74.54 ± 27.69	95.41 ± 6.75	90.39 ± 10.70	80.34 ± 12.23	73.01 ± 5.13	72.76 ± 16.54	80.71 ± 8.58	84.05 ± 6.54
TotalSeg. [6]	92.63 ± 11.19	90.40 ± 16.65	92.35 ± 7.18	67.05 ± 34.26	96.13 ± 5.64	90.46 ± 7.70	81.37 ± 7.16	70.04 ± 5.55	68.12 ± 19.32	79.20 ± 8.40	82.78 ± 6.82
WORD [45]	90.93 ± 9.43	86.81 ± 14.71	86.63 ± 17.43	66.20 ± 32.29	94.58 ± 5.06	86.20 ± 15.01	76.46 ± 11.75	68.33 ± 7.90	66.20 ± 16.27	78.67 ± 7.23	80.10 ± 6.21
Worst Selection	87.74 ± 16.12	84.15 ± 18.45	84.82 ± 19.22	59.59 ± 35.37	93.68 ± 7.23	84.24 ± 14.85	75.20 ± 11.80	66.17 ± 7.17	62.98 ± 19.11	73.18 ± 16.54	77.17 ± 8.99
Best Selection	94.85 ± 7.12	91.93 ± 13.62	93.40 ± 6.26	80.34 ± 23.91	96.39 ± 4.78	92.38 ± 7.34	83.81 ± 5.82	76.05 ± 4.45	75.44 ± 15.64	84.00 ± 6.52	86.86 ± 4.77
Random Selection	91.77 ± 15.07	89.13 ± 14.84	88.72 ± 17.59	67.70 ± 32.97	94.75 ± 7.34	87.43 ± 11.78	79.55 ± 10.89	70.53 ± 7.02	69.58 ± 16.19	79.56 ± 9.18	81.87 ± 6.14
Majority Voting Ensemble	93.62 ± 13.54	90.66 ± 13.14	90.26 ± 17.57	72.89 ± 30.82	95.81 ± 5.83	89.45 ± 11.71	80.84 ± 10.50	72.96 ± 5.26	71.41 ± 17.62	79.91 ± 8.90	83.78 ± 7.30
MLLM-guided Selection	92.68 ± 13.41	89.11 ± 16.59	89.43 ± 17.97	75.10 ± 27.27	95.54 ± 6.77	90.60 ± 10.20	80.92 ± 8.54	71.90 ± 4.93	71.37 ± 16.24	81.43 ± 7.41	83.81 ± 6.30
MLLM-guided Ensemble	92.27 ± 11.59	89.32 ± 15.67	89.20 ± 17.04	76.04 ± 27.44	95.69 ± 6.14	91.08 ± 10.09	82.20 ± 7.91	74.21 ± 4.25	73.82 ± 16.18	82.40 ± 6.73	84.63 ± 6.27

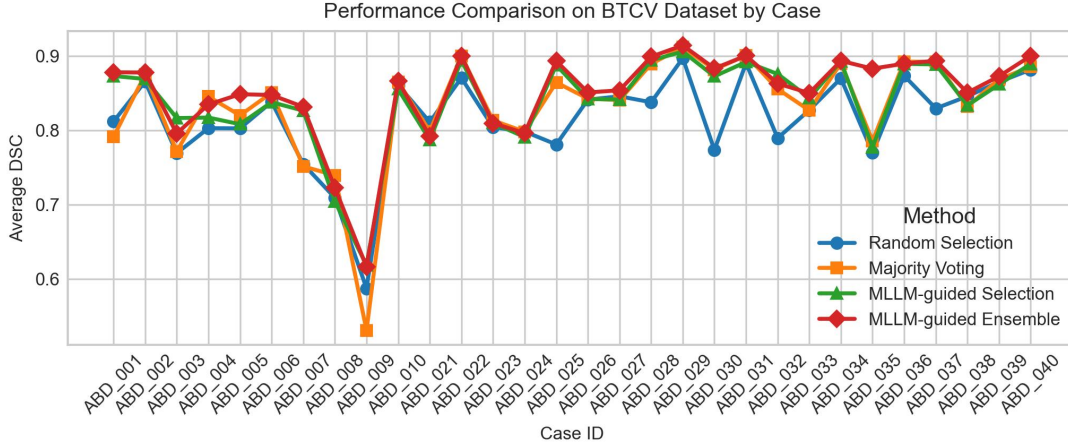


Fig. 7. Case-wise comparison of segmentation performance on the BTCV dataset using different results integration strategies, evaluated by average Dice Similarity Coefficient (DSC).

methods, MLLM-guided Ensemble achieved the highest DSC (84.63%), outperforming Majority Voting (83.78%, +0.85%), Random Selection (81.87%, +2.76%), and approaching the oracle upper bound (gap: 2.23%). MLLM-guided Selection (83.81%) also exceeded Random Selection (+1.94%), validating that visual comparison by MLLMs provides meaningful guidance. The superiority of MLLM-guided Ensemble over Selection (+0.82%) confirms the benefit of rank-aware voting in leveraging complementary model strengths.

V. DISCUSSION AND CONCLUSION

In this work, we present MedSegAgent, a multi-agent framework that orchestrates specialized segmentation models through natural language instructions. Rather than developing yet another universal model, our approach leverages existing task-specific models through intelligent selection and integration, achieving 94.27% complete coverage and 99.03% minimal coverage across 343 distinct segmentation queries spanning 23 datasets.

A core design choice is employing multiple specialized models rather than a single universal model. In our evaluated scenarios, specialized nnU-Net [15] models outperform prompt-based universal methods such as MedSAM2 [59], SegVol [10], and SAT-Pro [9] in both accuracy and efficiency (Tab. VI, Fig. 6). This strategy enhances scalability by enabling integration of new models without retraining (Tab. V). MedSegAgent aligns with the Mixture of Experts (MoE) [63] paradigm, dynamically invoking only the required models for

each query and loading only one or two compact models at inference time, requiring substantially less GPU memory than large universal models. The coarse-to-fine (C2F) matching strategy is critical for reliable semantic mapping: by progressively constraining the candidate space (modality → anatomy → label), it reduces semantic interference and mitigates failure modes such as dataset omission and label mismatch (Fig. 4, Tab. II). The hierarchical decomposition particularly benefits smaller LLMs by breaking complex matching into simpler sequential sub-tasks (Tab. III).

MedSegAgent can also integrate outputs from multiple models. As shown in Tab. VII, different models exhibit varied performance across clinical contexts. Recent work has demonstrated that MLLMs possess the capability to assess segmentation mask quality [64]; building on this insight, MedSegAgent employs MLLM-guided rank-aware ensemble to provide more robust results in complex cases. Practical considerations include potential quality misjudgment by MLLMs due to slice selection or visualization biases, and additional inference overhead from running multiple models.

The LLM component enables robust interpretation of clinical requirements and terminological variations (Fig. 5). While closed-source models like GPT-4o [11] may raise privacy concerns, locally deployable open-source models such as Qwen2.5 [13] achieve competitive performance (Tab. III). Regarding efficiency, cold start latency (40s for Qwen2.5-14B) represents a one-time model selection cost; once the model is selected, all subsequent inferences proceed without LLM

overhead. While this initial overhead may affect time-critical scenarios such as intraoperative guidance, it becomes amortized in typical clinical workflows where the same anatomical target is segmented across multiple patients or follow-up scans, making warm start the dominant scenario with negligible LLM overhead. For non-urgent tasks such as treatment planning or batch processing, the cold start overhead remains acceptable.

A practical trade-off exists across model scales: smaller models (7B-8B) enable faster deployment but show notable performance drops due to limited capacity, primarily affecting complex edge cases, while intermediate models like Qwen2.5-14B balance accuracy and deployability (Tab. III). This observation highlights a flexible deployment spectrum: MedSegAgent allows users to choose between high-performance mode (larger LLMs for maximum accuracy) and high-efficiency mode (smaller LLMs for faster response) depending on their computational constraints. Future work could train dedicated compact models optimized for medical semantic matching, offering better performance-efficiency trade-offs than general-purpose small LLMs.

Our approach has several limitations. First, MedSegAgent depends on the availability of suitable specialized models in its library. The segmentation accuracy in our comparisons (Tab. VI) reflects the underlying nnU-Net models; our contribution lies in the orchestration framework itself. Unlike universal models that can attempt any structure within their training distribution, our system requires a matching pre-trained model. This represents a fundamental trade-off between exploratory flexibility and clinical accuracy on known tasks. Second, while our benchmark systematically evaluates mapping accuracy using standard dataset naming conventions, real-world clinical queries may contain additional context or non-standard terminology. We partially address this through robustness evaluation with uncommon synonyms and medical abbreviations (Tab. IV), but future work should incorporate real clinician queries to assess end-to-end usability in practice.

In conclusion, by enabling clinicians to access specialized segmentation models through intuitive natural language queries, MedSegAgent demonstrates a practical pathway toward making advanced medical image analysis more accessible and scalable in clinical practice.

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REFERENCES

- [1] K. Cao *et al.*, “Large-scale pancreatic cancer detection via non-contrast CT and deep learning,” *Nat. Med.*, vol. 29, no. 12, pp. 3033–3043, 2023.
- [2] X. Luo *et al.*, “SegRap2023: a benchmark of organs-at-risk and gross tumor volume segmentation for radiotherapy planning of nasopharyngeal carcinoma,” *Med. Image Anal.*, vol. 101, p. 103447, 2025.
- [3] H. Wang *et al.*, “Video-instrument synergistic network for referring video instrument segmentation in robotic surgery,” *IEEE Trans. Med. Imaging*, vol. 43, no. 12, pp. 4457–4469, 2024.
- [4] M. Antonelli *et al.*, “The medical segmentation decathlon,” *Nat. Commun.*, vol. 13, no. 1, p. 4128, 2022.
- [5] J. Li *et al.*, “A systematic collection of medical image datasets for deep learning,” *ACM Comput. Surv.*, vol. 56, no. 5, pp. 116:1–116:51, 2023.
- [6] J. Wasserthal *et al.*, “TotalSegmentator: robust segmentation of 104 anatomic structures in CT images,” *Radiology: Artificial Intelligence*, vol. 5, no. 5, p. e230024, 2023.
- [7] T. A. D’Antonoli *et al.*, “TotalSegmentator MRI: robust sequence-independent segmentation of multiple anatomic structures in MRI,” *Radiology*, 2025.
- [8] J. Ma, Y. He, F. Li, L. Han, C. You, and B. Wang, “Segment anything in medical images,” *Nat. Commun.*, vol. 15, no. 1, p. 654, 2024.
- [9] Z. Zhao *et al.*, “Large-vocabulary segmentation for medical images with text prompts,” *npj Digit. Med.*, vol. 8, p. 566, 2025.
- [10] Y. Du, F. Bai, T. Huang, and B. Zhao, “SegVol: universal and interactive volumetric medical image segmentation,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024.
- [11] OpenAI, “GPT-4o system card,” *arXiv*, vol. abs/2410.21276, 2024.
- [12] Anthropic, “The claude 3 model family: Opus, sonnet, haiku,” 2024. [Online]. Available: <https://www-cdn.anthropic.com/de8ba9b01c9ab7cbabf5c33b80b7bbc618857627/Model.Card.Claude.3.pdf>
- [13] A. Yang *et al.*, “Qwen2.5 technical report,” *arXiv*, vol. abs/2412.15115, 2024.
- [14] O. Ronneberger, P. Fischer, and T. Brox, “U-Net: convolutional networks for biomedical image segmentation,” in *Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, 2015, pp. 234–241.
- [15] F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, “nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation,” *Nat. Methods*, vol. 18, no. 2, pp. 203–211, 2021.
- [16] Z. Zhou, M. M. R. Siddiquee, N. Tajbakhsh, and J. Liang, “UNet++: redesigning skip connections to exploit multiscale features in image segmentation,” *IEEE Trans. Med. Imaging*, vol. 39, no. 6, pp. 1856–1867, 2020.
- [17] Y. Tang *et al.*, “Self-supervised pre-training of Swin transformers for 3D medical image analysis,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 20 730–20 740.
- [18] M. Zhang *et al.*, “Multi-site, multi-domain airway tree modeling,” *Med. Image Anal.*, vol. 90, p. 102957, 2023.
- [19] Z. Huang *et al.*, “Revisiting model scaling with a U-net benchmark for 3D medical image segmentation,” *Sci. Rep.*, vol. 15, p. 29795, 2025.
- [20] J. Ma *et al.*, “Unleashing the strengths of unlabelled data in deep learning-assisted pan-cancer abdominal organ quantification: The FLARE22 challenge,” *The Lancet Digital Health*, vol. 6, no. 11, pp. e815–e826, 2024.
- [21] Y. Ji *et al.*, “AMOS: a large-scale abdominal multi-organ benchmark for versatile medical image segmentation,” in *Proc. NeurIPS Datasets Benchmarks Track*, 2022.
- [22] W. Li *et al.*, “AbdomenAtlas: A large-scale, detailed-annotated, & multi-center dataset for efficient transfer learning and open algorithmic benchmarking,” *Med. Image Anal.*, vol. 97, p. 103285, 2024.
- [23] J. Zhang, Y. Xie, Y. Xia, and C. Shen, “DoDNet: learning to segment multi-organ and tumors from multiple partially labeled datasets,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 1195–1204.
- [24] J. Liu *et al.*, “Universal and extensible language-vision models for organ segmentation and tumor detection from abdominal computed tomography,” *Med. Image Anal.*, vol. 97, p. 103226, 2024.
- [25] K. Singhal *et al.*, “Large language models encode clinical knowledge,” *Nature*, vol. 620, no. 7972, pp. 172–180, 2023.
- [26] Q. Jin *et al.*, “AgentMD: empowering language agents for risk prediction with large-scale clinical tool learning,” *Nat. Commun.*, vol. 16, p. 9377, 2025.
- [27] D. Van Veen *et al.*, “Adapted large language models can outperform medical experts in clinical text summarization,” *Nat. Med.*, vol. 30, no. 4, pp. 1134–1142, 2024.
- [28] I. Biri, U. T. Kucuktas, F. Uysal, and F. Hardalac, “Forecasting the future popularity of the anti-vax narrative on Twitter with machine learning,” *J. Supercomput.*, vol. 80, pp. 2917–2947, 2024.
- [29] C. Zakka *et al.*, “Almanac—retrieval-augmented language models for clinical medicine,” *NEJM AI*, 2024.
- [30] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, “Benchmarking retrieval-augmented generation for medicine,” in *Findings Assoc. Comput. Linguist. (ACL)*, 2024, pp. 6233–6251.

- [31] Y. Kim *et al.*, “MDAgents: an adaptive collaboration of LLMs for medical decision-making,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2024.
- [32] X. Tang *et al.*, “MedAgents: large language models as collaborators for zero-shot medical reasoning,” in *Findings Assoc. Comput. Linguist. (ACL)*, 2024, pp. 599–621.
- [33] B. Li *et al.*, “MMedAgent: learning to use medical tools with multi-modal agent,” in *Findings Assoc. Comput. Linguist. (EMNLP)*, 2024, pp. 8745–8760.
- [34] Z. Zhao *et al.*, “ChatCAD+: toward a universal and reliable interactive CAD using LLMs,” *IEEE Trans. Med. Imaging*, vol. 43, no. 11, pp. 3755–3766, 2024.
- [35] B. Rister, D. Yi, K. Shivakumar, T. Nobashi, and D. L. Rubin, “CT-ORG: a new dataset for multiple organ segmentation in computed tomography,” *Sci. Data*, vol. 7, no. 1, p. 381, 2020.
- [36] S. Gatidis *et al.*, “A whole-body FDG-PET/CT dataset with manually annotated tumor lesions,” *Sci. Data*, vol. 9, no. 1, p. 601, 2022.
- [37] U. Baid *et al.*, “The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification,” *arXiv*, vol. abs/2107.02314, 2021.
- [38] M. R. Hernandez Petzsche *et al.*, “ISLES 2022: a multi-center magnetic resonance imaging stroke lesion segmentation dataset,” *Sci. Data*, vol. 9, no. 1, p. 762, 2022.
- [39] X. Li *et al.*, “The state-of-the-art 3D anisotropic intracranial hemorrhage segmentation on non-contrast head CT: the INSTANCE challenge,” *arXiv*, vol. abs/2301.03281, 2023.
- [40] V. Oreiller *et al.*, “Head and neck tumor segmentation in PET/CT: the HECKTOR challenge,” *Med. Image Anal.*, vol. 77, p. 102336, 2022.
- [41] X. Zhuang, “Multivariate mixture model for myocardial segmentation combining multi-source images,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 41, no. 12, pp. 2933–2946, 2019.
- [42] O. Bernard *et al.*, “Deep learning techniques for automatic MRI cardiac multi-structures segmentation and diagnosis: is the problem solved?” *IEEE Trans. Med. Imaging*, vol. 37, no. 11, pp. 2514–2525, 2018.
- [43] A. Zeng *et al.*, “ImageCAS: a large-scale dataset and benchmark for coronary artery segmentation based on computed tomography angiography images,” *Comput. Med. Imaging Graph.*, vol. 109, p. 102287, 2023.
- [44] G. Luo *et al.*, “Efficient automatic segmentation for multi-level pulmonary arteries: the PARSE challenge,” *arXiv*, vol. abs/2304.03708, 2023.
- [45] X. Luo *et al.*, “WORD: a large-scale dataset, benchmark, and clinically applicable study for abdominal organ segmentation from CT images,” *Med. Image Anal.*, vol. 82, p. 102642, 2022.
- [46] N. Heller *et al.*, “The KiTS21 challenge: automatic segmentation of kidneys, renal tumors, and renal cysts in corticomedullary-phase CT,” *arXiv*, vol. abs/2307.01984, 2023.
- [47] P. Bilic *et al.*, “The liver tumor segmentation benchmark (LiTS),” *Med. Image Anal.*, vol. 84, p. 102680, 2023.
- [48] A. A. Ahmed *et al.*, “Radiomic mapping model for prediction of Ki-67 expression in adrenocortical carcinoma,” *Clin. Radiol.*, vol. 75, no. 6, pp. 479.e17–479.e22, 2020.
- [49] J. Wei *et al.*, “Chain-of-thought prompting elicits reasoning in large language models,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022, pp. 24 824–24 837.
- [50] S. Yao *et al.*, “ReAct: synergizing reasoning and acting in language models,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022.
- [51] P. Lewis *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 9459–9474.
- [52] A. Madaan *et al.*, “Self-Refine: iterative refinement with self-feedback,” in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 36, 2023, pp. 46 534–46 594.
- [53] A. Dubey *et al.*, “The Llama 3 herd of models,” *arXiv*, vol. abs/2407.21783, 2024.
- [54] Q. Wu *et al.*, “AutoGen: enabling next-gen LLM applications via multi-agent conversation,” *arXiv*, vol. abs/2308.08155, 2023.
- [55] R. Dorent *et al.*, “LNQ 2023 challenge: benchmark of weakly-supervised techniques for mediastinal lymph node quantification,” *Machine Learning for Biomedical Imaging*, vol. 3, pp. 1–15, 2025.
- [56] M. Masoudi, H.-R. Pourreza, M. Saadatmand-Tarzjan, N. Eftekhari, F. S. Zargar, and M. P. Rad, “A new dataset of computed-tomography angiography images for computer-aided detection of pulmonary embolism,” *Sci. Data*, vol. 5, no. 1, p. 180180, 2018.
- [57] M. A. D. Buser *et al.*, “Surgical planning in pediatric neuroblastoma,” in *Proc. Med. Image Comput. Comput.-Assist. Interv. (MICCAI)*, 2023.
- [58] R. Dorent *et al.*, “CrossMoDA 2021 challenge: benchmark of cross-modality domain adaptation techniques for vestibular schwannoma and cochlea segmentation,” *Med. Image Anal.*, vol. 83, p. 102628, 2023.
- [59] J. Ma *et al.*, “MedSAM2: segment anything in 3D medical images and videos,” *arXiv*, vol. abs/2504.03600, 2025.
- [60] H. Wang *et al.*, “SAM-Med3D: a vision foundation model for general-purpose segmentation on volumetric medical images,” *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 10, pp. 17 599–17 612, Oct. 2025.
- [61] B. Landman, Z. Xu, J. Iglesias, M. Styner, T. Langerak, and A. Klein, “MICCAI multi-atlas labeling beyond the cranial vault: workshop and challenge,” in *Proc. MICCAI Workshop Multi-Atlas Labeling Beyond Cranial Vault*, vol. 5, 2015, p. 12.
- [62] Z. Huang *et al.*, “A-Eval: a benchmark for cross-dataset and cross-modality evaluation of abdominal multi-organ segmentation,” *Med. Image Anal.*, vol. 101, p. 103499, 2025.
- [63] R. A. Jacobs, M. I. Jordan, S. J. Nowlan, and G. E. Hinton, “Adaptive mixtures of local experts,” *Neural Comput.*, vol. 3, no. 1, pp. 79–87, 1991.
- [64] P. R. A. S. Bassi *et al.*, “Label critic: Design data before models,” in *Proc. IEEE Int. Symp. Biomed. Imaging (ISBI)*, 2025.